B.Sc Forensic Science – Forensic Anthropology &Odontology(FS3401)

PART A

Q1. Intramembranous ossification starts after ___ month
Answer: 2nd month of embryonic development

Q2. ___ is the longest bone of human body
□ **Answer:** Femur

Q3. ___ helps in determination of age from skull
□ **Answer:** Closing of suture

Q4. Overall size of skull is typically larger in ___
□ **Answer:** Male skull

Q5. Study of somatoscopy can help in ___
□ **Answer:** Identifying individuals

Q6. Cranial index is used to classify
□ **Answer:** Head shape

Q7. Bertillonage is a system of identification based on
□ **Answer:** Body measurements

Q8. Tissue depth is important in facial reconstruction because it
□ **Answer:** Determines nose shape, eye position and facial appearance

Q9. Gustafson's method is used to estimate
□ **Answer:** Age at death

Q10. Dental records are highly reliable because
□ **Answer:** They are unique to each individual

PART B

Q11.1 Process of Endochondral Ossification

Endochondral ossification is the process by which long bones develop from a cartilage model. Initially, mesenchymal cells differentiate into chondrocytes forming a hyaline cartilage model. The cartilage grows in length and width. A primary ossification center appears in the diaphysis where blood vessels invade and osteoblasts begin bone formation. Cartilage cells degenerate and are replaced by bone tissue. Secondary ossification centers develop in the epiphyses after birth. This process continues until epiphyseal plates close, resulting in complete bone growth.

Q11.2 Significance of Anthropology in Forensic Science

Forensic anthropology applies the knowledge of human skeletal biology to medico-legal investigations. It helps in determining age, sex, stature, ancestry, and identification of skeletal remains. This discipline is especially useful in cases involving decomposed, burnt, or skeletonized bodies. It also assists in mass disaster victim identification and reconstruction of biological profile, aiding law enforcement agencies in solving crimes.

Q12.1 Characteristic Features of Male Skull

The male skull is generally larger, heavier, and more robust than the female skull. It has prominent supra-orbital ridges, a sloping forehead, large mastoid processes, thick cranial bones, and a square chin. The occipital protuberance is well developed. These features help forensic experts determine sex from skeletal remains.

Q12.2 Characteristic Features of Mandible

The mandible is the strongest bone of the face. It consists of a body and two rami. The male mandible is larger and heavier with a square chin and everted gonial angle, whereas the female mandible is smaller with a rounded chin and obtuse gonial angle. The mental foramen, mandibular condyle, and angle of mandible are important landmarks used in forensic identification.

Q13.1 Forensic Importance of Darwin's Tubercle, Ear Lobes, Supra-orbital Ridges and Scars

Darwin's tubercle is a congenital trait seen on the ear and helps in individual identification. Ear lobes (attached or free) are useful hereditary markers. Supra-orbital ridges show sexual dimorphism. Scars and occupational marks are acquired characteristics and provide strong

corroborative evidence in personal identification. However, these features should be used along with other identification methods due to variability.

Q13.2 Somatometry and Its Forensic Significance

Somatometry involves the measurement of different body parts such as head, face, nose, ear, and height. These measurements help in estimating stature and identifying individuals in medico-legal cases. Population-specific data increases accuracy. Somatometry is useful in living persons, skeletal remains, and mass disaster investigations when other methods are unavailable.

PART C

Q16.1 Comparative Skeletal Anatomy of Human and Non-Human Bones

Comparative skeletal anatomy is the study of differences between human and animal bones. Human bones are adapted for bipedal locomotion, while non-human bones are adapted for quadrupedal movement. The human skull has a large, rounded cranial vault with a high forehead and centrally placed foramen magnum, whereas animal skulls are elongated with posteriorly placed foramen magnum. The human femur shows a bicondylar angle, which is absent in animals. Human vertebral column is S-shaped, while animals have a straight or slightly curved spine. These features help forensic experts differentiate human skeletal remains from non-human bones.

OR

Q16.2 Human Skeletal System

The human skeletal system consists of 206 bones divided into axial and appendicular skeleton. The axial skeleton includes skull, vertebral column, ribs, and sternum, providing protection to vital organs. The appendicular skeleton includes bones of upper and lower limbs, enabling movement. Bones provide support, protection, movement, mineral storage, and blood cell formation. A well-labelled diagram includes skull, vertebrae, ribs, pelvis, femur, tibia, humerus, radius, and ulna.

Q17.1 Differences Between Male and Female Skull

The male skull is larger, heavier, and more robust than the female skull. Male skull shows prominent supra-orbital ridges, sloping forehead, large mastoid processes, thick cranial bones, square chin, and pronounced occipital protuberance. Female skull is smaller and smoother with sharp supra-orbital margins, vertical forehead, small mastoid processes, thin bones, rounded chin, and less prominent muscle attachments. These differences are used in forensic anthropology to determine sex with high accuracy.

OR

Q17.2 Male and Female Skull Diagrams

Male skull diagram shows prominent brow ridges, sloping forehead, square chin, and large mastoid process. Female skull diagram shows smooth brow ridges, vertical forehead, rounded chin, and small mastoid process. These visual differences help in easy sex determination.

Q18.1 Somatometric Measurements in Personal Identification

Somatometric measurements include head length, facial width, nasal height, ear length, stature, and body weight. These measurements help in personal identification and biological profiling. While stature estimation is highly reliable, facial and nasal measurements show population variability. Environmental, genetic, and nutritional factors influence accuracy. When combined with other forensic methods, somatometry plays an important role in identification across diverse populations.

OR

Q18.2 Cephalic, Nasal, Cranial and Upper Facial Indices

Cephalic index is calculated as head breadth × 100 / head length and helps classify head shape. Nasal index is nasal width × 100 / nasal height and helps in population affinity estimation. Cranial and upper facial indices assist in understanding anatomical variation and ancestry. These indices are useful in forensic anthropology to relate skeletal features to ethnic or regional background.

Q19.1 Facial Superimposition: Strengths and Limitations

Facial superimposition is a technique where skull images are superimposed on antemortem photographs. Strengths include non-destructive nature, low cost, and usefulness in exclusion. Limitations include dependence on photo quality, pose, and subjective interpretation. It is mainly used as corroborative evidence and not as sole proof of identity.

OR

Q19.2 Workflow of Facial Reconstruction

Facial reconstruction starts with skull examination and assessment of age, sex, and ancestry. Tissue depth markers are placed based on population data. Muscles are reconstructed followed by facial features like nose, lips, and eyes. Final facial approximation is created using clay or digital methods. Each stage helps narrow down identity in unknown cases.

Q20.1 Role of Forensic Odontology in Mass Disasters

Forensic odontology plays a vital role in mass disasters when bodies are mutilated or decomposed. Teeth are resistant to fire and trauma. Antemortem dental records are compared with postmortem findings for identification. It is cost-effective, accurate, and widely accepted in courts when other identification methods fail.

OR

Q20.2 Bite Mark Evidence in Personal Identification

Bite mark evidence involves comparison of bite patterns with dentition of suspects. It can link an individual to a crime. However, skin distortion and healing can affect accuracy. Proper photographic documentation and expert analysis are required. Bite mark evidence is admissible in court when supported by forensic protocols and corroborative evidence.